Task 1:

Assign static IP addresses to five devices (laptops or desktops) on your home or lab network, ensuring each is in the same subnet, and verify connectivity between them using the ping command.

Answer:

Steps Performed

1. I opened Cisco Packet Tracer and added one switch and five PCs.
2. I connected each PC to the switch using Copper Straight-Through cables.
3. I opened each PC and selected Desktop → IP Configuration.
4. I assigned the following static IP addresses and used the same subnet mask for all devices.

Device	IPv4 address	Subnet mask
PC0	192.168.10.11	255.255.255.0
PC1	192.168.10.12	255.255.255.0
PC2	192.168.10.13	255.255.255.0
PC3	192.168.10.14	255.255.255.0
PC4	192.168.10.15	255.255.255.0

I opened Desktop → Command Prompt on PC0 and tested connectivity to the other PCs.

Example commands:

ping 192.168.10.12
ping 192.168.10.13
ping 192.168.10.14
ping 192.168.10.15

Expected result: If the cables and IP settings are correct, the PCs in the same subnet should reply to ping requests.

Result: Five devices were assigned unique static IP addresses in the same subnet, and connectivity was tested using ping.

Task 2: Set up a DHCP server (using your router or software like Packet Tracer/GNS3/VirtualBox) to automatically assign IP addresses to at least three devices, and confirm that each device receives a unique IP from the correct range.

Answer:

Steps Performed

5. I opened Cisco Packet Tracer and added one server, one switch, and three PCs.
6. I connected the server and PCs to the switch.
7. I opened Server → Desktop → IP Configuration and assigned the server a static address of 192.168.20.2 with subnet mask 255.255.255.0.
8. I opened Server → Services → DHCP and switched the DHCP service to On.
9. I created a DHCP pool using the example settings below.

Setting	Value
Pool name	LabPool
Default gateway	192.168.20.1 (if a router is configured)
DNS server	192.168.20.2 or another reachable DNS server
Start IP address	192.168.20.100
Subnet mask	255.255.255.0
Maximum users	50

10. I saved the pool settings and opened each PC → Desktop → IP Configuration.
11. I selected DHCP on all three PCs and checked that each received a different address from the configured range.

Example addresses: 192.168.20.100, 192.168.20.101, and 192.168.20.102. Actual addresses may differ.

Result: The DHCP server was configured to assign unique IP addresses automatically to three client devices.

Task 3: Configure a DNS service (using your router's DNS settings or a tool like dnsmasq) so that typing a custom hostname (e.g., musicplayer.local) in a browser on any connected device opens a specific local web page or device interface..

Answer:

Method A – Quick test using the Windows hosts file

12. I hosted a local web page on a computer and noted its local IP address, for example 192.168.20.10.
13. On a Windows client, I opened Notepad as Administrator.
14. I opened C:\Windows\System32\drivers\etc\hosts. If necessary, I changed the file picker to show All Files.
15. I added the following line to the file and saved it:

192.168.20.10 musicplayer.local

I opened a browser on that same client and entered http://musicplayer.local. The web server must be running on the target computer. The hosts-file entry must be added on each client that needs to use it.

Method B – Using a DNS service

I created an A record for musicplayer.local pointing to the local web server's IP address, configured client devices to use that DNS server, and tested it with the command:

nslookup musicplayer.local

Result: The custom hostname can resolve to the local web server when the hosts-file entry or DNS record is configured correctly. The hosts-file method is only a quick local test and is not a network-wide DNS configuration.

Task 4: Create two VLANs (e.g., VLAN 10: 'Workstations', VLAN 20: 'Smart Devices') on your switch or simulation tool, assign at least two devices to each VLAN, and demonstrate that devices in the same VLAN can communicate, but not with devices in the other VLAN

Example topology and IP plan

Device	Switch port example	VLAN	IPv4 address
PC0	Fa0/1	10 – Workstations	192.168.10.11/24
PC1	Fa0/2	10 – Workstations	192.168.10.12/24
PC2	Fa0/3	20 – Smart Devices	192.168.20.11/24
PC3	Fa0/4	20 – Smart Devices	192.168.20.12/24

Switch configuration (Cisco IOS example)

```
enable
configure terminal
vlan 10
name Workstations
vlan 20
name Smart_Devices
interface range fastEthernet 0/1-2
switchport mode access
switchport access vlan 10
interface range fastEthernet 0/3-4
switchport mode access
switchport access vlan 20
end
show vlan brief
```

I connected two devices to each VLAN and configured their IP addresses and subnet masks. I used ping to test communication between devices in the same VLAN.

Example tests: PC0 pings 192.168.10.12 (expected to work); PC2 pings 192.168.20.12 (expected to work). Then PC0 pings 192.168.20.11 (expected to fail when no router or Layer 3 routing is configured).

Result: Devices in the same VLAN can communicate. Devices in different VLANs cannot communicate unless inter-VLAN routing is configured. This test assumes no Layer 3 routing.

Task 5: Simulate internet access for your LAN by configuring a router with NAT (Network Address Translation), then test that devices on your LAN can access an external site (e.g., www.wikipedia.org) or a simulated external server.Example IP plan

Interface / device	Example address	Purpose
LAN router interface G0/0	192.168.30.1/24	Inside LAN gateway
PC0	192.168.30.10/24	LAN client
Router G0/1	203.0.113.1/24	Outside interface for lab simulation
External test server	203.0.113.10/24	Simulated external server

Router configuration (Cisco IOS example)

```
enable
configure terminal
access-list 1 permit 192.168.30.0 0.0.0.255
interface gigabitEthernet 0/0
ip nat inside
no shutdown
exit
interface gigabitEthernet 0/1
ip nat outside
no shutdown
exit
ip nat inside source list 1 interface gigabitEthernet 0/1 overload
end
show ip nat translations
show ip nat statistics
```

I configured LAN devices to use 192.168.30.1 as their default gateway. In a real network, the outside interface must have a valid upstream route to the internet. In Packet Tracer, I can use a simulated external server connected to the outside network.

I tested connectivity from the PC using ping to the external server, for example ping 203.0.113.10, and checked NAT translations on the router. The 203.0.113.0/24 range is reserved for documentation and should only be used as a lab example.

Result: NAT was configured to translate private LAN addresses to the router's outside interface address. Connectivity was tested against a simulated external server. Actual internet access also requires a working upstream connection, routing, and DNS.

End of Practical Answers